

UK Student Visa and Work Rights Advisor

A Rule-Based Conversational AI Agent

CMP-N206-0 Artificial Intelligence | Coursework 2

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1. Domain Description and Problem Statement

International students in the United Kingdom navigate a complex legal landscape where a single misunderstanding of work conditions can have life-altering consequences. Working even a few hours over the permitted limit during term time constitutes a breach of visa conditions that can result in immediate removal from the country and a ten-year re-entry ban [3]. While official GOV.UK guidance is comprehensive, it is dense and difficult to interpret under the pressure of a busy academic term. The UK Council for International Student Affairs (UKCISA) notes that confusion around term-time and holiday periods is one of the most frequently reported issues amongst international students [4].

I built this agent to solve a specific, high-stakes problem: providing immediate, accurate, and traceable advice on work rights under the UK Student visa. By focusing on a narrow and factually grounded domain, I have prioritised legal reliability over the unpredictable generative behaviour of large language models. This agent acts as a safety-first tool, helping students remain compliant with the law while they continue their studies. The system follows the PEAS framework described by Russell and Norvig [1]: it senses user text as its percept, processes it through structured legal logic, and produces signposting responses or risk warnings as its actions.

1.1 PEAS Model

Component	Definition in this Agent
Performance	Accuracy of guidance given, breach risk detection rate, appropriate uncertainty handling, clarity of explanations
Environment	UKVI Student visa regulations [3]; UK academic calendar conventions; UKCISA guidance [4]
Actuators	Natural language text output: answer a query, request clarification, issue a risk warning, or refuse an unsafe request
Sensors	User natural language input, tokenised and classified at runtime through the Bayesian intent pipeline

2. Knowledge Representation

The agent uses two complementary knowledge representation strategies, both of which are identified by Russell and Norvig [1] as standard approaches for rule-based intelligent systems.

2.1 Fact Database

A structured Python dictionary encodes ground-truth UKVI rules sourced from official GOV.UK documentation [3] and UKCISA guidance [4]. This constitutes propositional knowledge about the domain: the 20-hour weekly term-time ceiling, the absolute prohibition on self-employment and company directorships, the absence of a weekly cap during official university holidays, and the consequences of a breach. Each fact references its official source so that the knowledge base can be audited and updated when UKVI policy changes.

2.2 Production Rule Base

Seven IF-THEN production rules encode conditional knowledge about the domain. Each rule contains: a unique rule identifier, a keyword set for matching, a natural language response drawn directly from official guidance, an explanation string for the explainability layer, and a domain classification. An example rule is as follows:

IF message contains { "freelance", "self-employ", "sole trader", "own business" }

THEN fire Rule R02: self-employment is prohibited on a UK Student visa regardless of hours.

This structure ensures every agent decision is traceable back to a specific, citable source. Adadi and Berrada [16] identify traceability as a foundational requirement for explainable AI systems operating in high-stakes domains, and Chazette et al. [17] demonstrate that rule-based traceability directly supports transparency as a software quality requirement.

3. Inference Method

The agent uses a dual-layer inference architecture. Both layers run on every query and their outputs are combined in the final response. Figure 1 below illustrates the full query processing pipeline from user input to final output.

Figure 1: Agent Query Processing Pipeline



3.1 Bayesian Intent Classification

To handle the natural variability of human language, I implemented a Naive Bayes classifier [8][10]. When a student asks about picking up extra shifts or working over the holidays, the classifier calculates the posterior probability of their intent matching each of the seven rules. The Naive Bayes approach is justified by its strong empirical performance on text classification tasks with limited training vocabulary, as demonstrated by Rish [8] and Jiang and Lee [9]. Zhang [10] further establishes the theoretical optimality of Naive Bayes classifiers under certain distributional assumptions, lending formal backing to the approach.

$$P(\text{intent}_i | \text{message}) \propto P(\text{message} | \text{intent}_i) \times P(\text{intent}_i)$$

A uniform prior is assigned across all seven intents before any evidence is observed. The likelihood term is computed as a Laplace-smoothed keyword overlap score [11][12]. Laplace smoothing with parameter $\alpha = 0.5$ (Jeffreys prior) prevents zero-probability estimates when the user's vocabulary falls outside the keyword set, which MacCartney [12] identifies as the fundamental motivation for additive smoothing in probabilistic classifiers. After computing unnormalised posteriors for all intents, the distribution is normalised to sum to one. A confidence threshold of 0.22 gates inference: if no intent exceeds this posterior, the agent requests clarification rather than acting on weak evidence.

3.2 Forward-Chaining Risk Assessment

In parallel, a forward-chaining inference engine [1][13] evaluates six risk factors derived from the available evidence in the user's message. This data-driven approach starts from known facts and works towards a conclusion, which is the defining characteristic of forward chaining as described by Mzori and Abdulla [13] and implemented in expert systems by Kapoor et al. [14]. Starting from a base prior of 5%, multiplicative risk factors are applied sequentially as each factor fires:

Risk Factor	Multiplier	Rationale
Hours reported above 20	×15.0	Direct evidence of a limit breach; the most significant signal available
Hours between 18 and 20	×2.5	Proximity to the limit substantially raises the probability of accidental breach
Self-employment mentioned	×12.0	Categorically prohibited under Student visa conditions regardless of hours
Term time period confirmed	×3.0	Confirms the 20-hour limit is applicable to the user's situation
Multiple employers mentioned	×1.8	Increases the likelihood of miscounting combined hours across employers
Breach language used by user	×5.0	User's own phrasing suggests awareness of a potential compliance issue

The final score is clamped to the interval [0, 1] and mapped to four levels: Low (below 15%), Moderate (15 to 45%), High (45 to 75%), and Critical (above 75%). Risk warnings are only surfaced to the user when the score exceeds 20%, avoiding alert fatigue for routine informational queries.

4. Uncertainty Handling

Handling uncertainty responsibly is particularly important in this domain: an overconfident agent that gives incorrect immigration advice could cause direct harm to a student. The agent addresses uncertainty at three distinct levels.

- Laplace smoothing: Applied with $\alpha = 0.5$ to prevent zero-probability estimates in the Bayesian classifier. Manning et al. [11] and MacCartney [12] both establish this as the standard approach for handling unseen vocabulary in probabilistic text classifiers, and it ensures the agent degrades gracefully when encountering informal or unfamiliar phrasing.
- Confidence thresholding: If no rule achieves a posterior probability of 0.22 or above, the agent does not act on its best guess. Instead, it selects from a rotating set of clarification prompts, guiding the user towards a more parseable query. This prevents potentially dangerous incorrect advice being given on the basis of weak or ambiguous evidence.
- Probabilistic risk scoring: Rather than issuing a binary compliant or in breach judgement, the engine produces a continuous risk score mapped to four human-readable levels. This communicates graduated uncertainty to the user and avoids the false confidence of categorical outputs in situations where the evidence is mixed.
- Transparent output: Every response includes a full ranked posterior distribution across all seven intents, displaying the agent's confidence levels explicitly. Arrieta et al. [18] and Chazette et al. [17] both identify this form of output-level explainability as critical for building user trust in high-stakes advisory systems.

5. Ethical Reflection

5.1 Bias and Fairness

The agent applies UKVI rules uniformly across all users, which avoids direct discrimination. However, a meaningful bias risk exists through omission: the system is scoped to the Student visa only, and users on Graduate visas, Skilled Worker visas, or Dependant visas may receive inapplicable guidance. Mehrabi et al. [20] identify this as representation bias, a systemic concern in AI systems that serve heterogeneous populations. A future version should include a visa type elicitation step at the start of every conversation, and should explicitly decline queries that fall outside its validated knowledge scope.

5.2 Misinformation Risk

Immigration rules are subject to policy change without notice. An agent providing outdated information could cause a student to breach their visa conditions, an outcome far more damaging than receiving no advice at all. This risk is mitigated through three design decisions: (a) explicit disclaimers appear on every response; (b) every rule response cites its official GOV.UK source [3]; and (c) every response redirects the user to their university's international student support team.

The agent is positioned as a signposting tool rather than a legal authority, in line with Hagendorff's principle that AI systems should clearly demarcate their competence boundaries [19].

5.3 Safety by Design

A regex-based safety filter intercepts and refuses any request that appears to seek advice on evading visa conditions. This is a non-negotiable boundary: the agent will not facilitate immigration fraud regardless of how the request is framed. When a refusal is triggered, the agent explains why and redirects to legitimate institutional support. Fjeld et al. [19] map this design property to the internationally recognised AI ethics principle of non-maleficence, which holds that AI systems must be designed not to cause harm to their users or to third parties.

5.4 Sustainability

I chose this symbolic AI approach in part because it is far more sustainable than using generative language models. This system runs entirely on local processing power with no API calls and negligible energy consumption. Strubell et al. [5] demonstrated that training and running large neural language models incurs substantial carbon costs that are entirely avoidable when the problem is tractable with classical methods. Schwartz et al. [6] extend this argument in their Green AI framework, calling on researchers to report and minimise computational costs as a first-class concern. For a domain as well-defined as UK Student visa work rights, a rule-based symbolic system delivers superior reliability, interpretability, and energy efficiency.

5.5 Alignment with the EU AI Act

Under the EU AI Act [2], this agent would be classified as a low-risk AI system. It does not make binding administrative decisions, does not process sensitive personal data, operates with fully traceable rule-based logic, and includes explicit human oversight recommendations in every response. The explainability layer directly addresses the transparency requirements set out in Article 13 of the Act. Fjeld et al. [19] map these design properties to the broad international consensus around transparency, accountability, and non-maleficence in AI development.

6. Evaluation

6.1 Success Cases

Scenario	Input Example	Agent Outcome
Standard hours query	How many hours can I work per week?	Rule R01 fired at 98.2% confidence. Correct 20-hour limit stated with holiday distinction included.

Confirmed breach	I worked 25 hours during term time last week	Rule R05 fired. Risk score: Critical (100%). Consequences stated; university support signposted.
Self-employment query	Can I do some freelance web development on the side?	Rule R02 fired at 97.8% confidence. Prohibition stated clearly with source citation.
Holiday work	It is summer break, can I work full time?	Rule R04 fired at 83.1% confidence. Full-time confirmed; resit period caveat included.
Multiple employers	I have two part-time jobs during term time	Rule R07 fired. Combined-hour aggregation rule explained. Risk: Moderate.

6.2 Failure Cases and Limitations

Failure	Root Cause	Proposed Improvement
Slang not recognised (e.g. "graft a bit extra")	Keyword set lacks informal vocabulary	Integrate word embeddings such as spaCy for semantic similarity [15]
Granular edge cases not handled	No hour accumulation logic across turns	Session state tracking to aggregate hours across multiple conversational turns
No memory between turns	Stateless architecture by design	Dialogue state manager to enable multi-turn contextual reasoning [15]
Knowledge base may become outdated	Rules are static and require manual review	Version-tagged rules with last-verified dates and automated staleness alerts
Scoped to Student visa only	Deliberate narrow domain choice	Visa type elicitation step; extended rule base covering Graduate and Dependant visas

7. Conclusion

This project demonstrates that artificial intelligence is at its most meaningful when it is applied to protect and empower people in genuinely vulnerable situations. By combining the probabilistic strengths of Bayesian reasoning [8][10] with the rigorous traceability of a rule-based inference engine [13][14], I have built a system that provides actionable, safe, and transparent guidance for international students navigating the UK visa system.

The limitations identified in Section 6 are genuine rather than cosmetic. Addressing them through semantic NLP integration [15], multi-turn dialogue state management, and an expanded visa coverage model would substantially improve the agent's real-world utility. The ethical analysis in Section 5 situates the system within the broader landscape of responsible AI design [16][17][18][19][20], demonstrating that even a rule-based system of this scale requires careful attention to fairness, transparency, and the long-term welfare of its users.

This agent was built with the belief that clear, trustworthy, and energy-efficient AI can solve real problems without the complexity or the carbon cost of large-scale generative models [5][6]. For a problem as well-defined and as consequential as this one, a classical symbolic approach is not a compromise. It is the right choice.

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